

Zizhe Zhang

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EDUCATION

Master of Science in Engineering (M.S.E.) in Robotics

University of Pennsylvania, GRASP Lab

Advisor: Prof. Nadia Figueroa

Aug 2024 – Present

Philadelphia, PA

Visiting Student

University of California San Diego, Jacobs School of Engineering

Jan 2023 – Mar 2023

San Diego, CA

Bachelor of Engineering (B.E.) in Measuring Control Technology & Instruments

Southeast University, School of Instrument Science and Engineering

Advisor: Prof. Yuan Yang

Aug 2020 – Jun 2024

Nanjing, China

PUBLICATIONS

(* and † denote equal contribution)

Viability-Preserving Passive Torque Control

Zizhe Zhang*, Yicong Wang*, Zhiquan Zhang, Tianyu Li, and Nadia Figueroa

Under Review

[[Website](#)]

Flow with the Force Field: Learning 3D Compliant Flow Matching Policies from Force and Demonstration-Guided Simulation Data

Tianyu Li*, Yihan Li*, Zizhe Zhang, and Nadia Figueroa

Under Review

[[Website](#)]

VLMgineer: Vision Language Models as Robotic Toolsmiths

George Jiayuan Gao*, Tianyu Li*, Junyao Shi, Zizhe Zhang†, Yihan Li†, Nadia Figueroa, and Dinesh Jayaraman

Under Review

[[arXiv](#)] [[PDF](#)] [[Website](#)]

Robotics: Science and Systems (RSS) Workshop on Robot Hardware-Aware Intelligence (**Oral Spotlight**), 2025

Image-Based Visual Servoing for Enhanced Cooperation of Dual-Arm Manipulation

Zizhe Zhang, Yuan Yang, Wenqiang Zuo, Guangming Song, Aiguo Song, and Yang Shi

IEEE Robotics and Automation Letters (RA-L), 2025

[[arXiv](#)] [[PDF](#)] [[Website](#)]

RESEARCH AND INDUSTRY EXPERIENCE

Visiting Scholar, Duke University

Robot Dexterity Lab, advised by Prof. Xianyi Cheng

Working on developing an adaptive compliance motion representation for generalizable robotic manipulation and physical human-robot interaction (In Progress)

Durham, NC

Jun 2025 – Aug 2025

Graduate Research Assistant, University of Pennsylvania

Figueroa Robotics Lab, advised by Prof. Nadia Figueroa

Working on machine learning applications in control theory and robot manipulation. Projects include:

- Ensuring feasibility of passivity-based torque control via learned kinematic constraint mappings (*Under Review*)
- Implementing vision-based learning with force feedback for contact-rich manipulation tasks (*Under Review*)
- Utilizing Vision Language Models (VLMs) for robot policy synthesis ([RSS Workshop 2025](#))

Philadelphia, PA

Nov 2024 – Present

Technical Intern, Schneider Electric

Working in the Kylin project, including:

- IGBT thermal simulations
- Capacitor lifetime assessments
- EMC testing
- Circuit design

Shanghai, China

Jun 2023 – Aug 2023

Undergraduate Research Assistant, Southeast University

Robotic Perception and Control Lab, advised by Prof. Yuan Yang

Worked on developing visual-servoing-based robotic control architectures. Projects include:

Designed an enhanced dual-arm collaborative control system based on image-based visual servoing (RA-L 2025)

Designed a shared teleoperation system control based on visual servoing (Bachelor Thesis)

Nanjing, China

Dec 2023 – Aug 2024

Automotive Safety Technology Lab, advised by Prof. Dong Wang

Applying edge detection to classify Martian topography and identify soft-ground hazards

Designed a wheeled-leg ground-detection mechanism for rover traversal and analyzing force signals to predict mobility

Mar 2022 – Jun 2023

Advanced Navigation Technology Lab, advised by Prof. Xuanpeng Li

Utilizing the PYTS library for signal visualization and building a CNN to extract and classify features from ADS-B radio signals

Jul 2023 – Sep 2024

COURSE PROJECTS

News Source Classification, University of Pennsylvania

Team of 3, Leader

Collected and preprocessed 415,343 headlines from Fox News and NBC using sitemap scraping and BeautifulSoup, ensuring a balanced, cleaned dataset

Built a TF-IDF + Logistic Regression baseline and fine-tuned BERT and DeBERTa models, applying k-fold cross-validation to eliminate data leakage

DeBERTa K-fold achieved 88.66 % test accuracy, significantly surpassing the baseline (67.61 %)

Philadelphia, PA

Aug 2024 – Dec 2024

Autonomous Vehicle based on GPS & DoF Camera, University of California San Diego

Team of 3

Utilized Python and VESC to control the robot, DoF camera to find and track lanes, centimetric GPS and PID method to record and follow paths

Brought the robot to a complete stop by using PyVesc and DepthAI libraries to run stop sign detection on the camera

Enabled the robot to respond correctly to speed limit signs by performing text detection on the camera

San Diego, CA

Jan 2023 – Mar 2023

Design and Implementation of a Weather Query and Display Module, Southeast University.

Team of 3, Leader

Used ASM to design the function and organize each part of the MCU. Designed a Weather Query and Display Module which can choose and display weather information of several customized cities

The whole project is based on a small device which can be used as smart home device

Nanjing, China

Jan 2023 – Mar 2023

HONORS AND AWARDS

Outstanding Bachelor’s Thesis, Southeast University

Merit Student, Southeast University

2024

2021

SERVICE

Reviewer

IEEE Transactions on Robotics (TRO)

IEEE Transactions on Industrial Electronics (TIE)

IEEE Transactions on Industrial Informatics (TII)

2025

2025

2025

Teaching Assistant

ESE 5000 Linear Systems Theory, University of Pennsylvania

Fall 2025

Volunteer

Robotics: Science and Systems (RSS)

2025

SKILLS

Programming

Tools

Robots

C, C++, Python, MATLAB

CMake, PyTorch, ROS, RLBench, ManiSkill, CoppeliaSim, PyBullet, Issac Gym/Sim/Lab

UR3, UR3e, Franka Emika Panda, Franka Research 3